

Claude

Parameter	Value
Input voltage (LiPo 2S battery)	VIN = 7.4 V
Output voltage	VOUT = 3.3 V
Nominal output current	IS = 200 mA
Inductor current ripple	$\Delta IL = 30\%$ of IS
Max output voltage ripple	$\Delta VS = 50$ mV
Switching frequency (LT1959)	f = 500 kHz

1. Duty Cycle

For an ideal buck converter in continuous mode:

$$D = V_{OUT} / V_{IN}$$

$$D = 3,3 / 7,4$$

$$D = 0,446 \rightarrow t_{on} = 0,892 \mu s / t_{off} = 1,108 \mu s$$

Check : $D = 44,6 \% < 50 \% \rightarrow$ peak current $I_P = 4,5$ A guaranteed by LT1959.

2. Inductor L1

2.1 Target current ripple

The peak-to-peak current ripple in the inductor is set to 30% of the output current:

$$\Delta IL = 0,30 \times IS = 0,30 \times 200 \text{ mA}$$

$$\Delta IL = 60 \text{ mA}$$

2.2 Minimum value of L

The formula from the datasheet LT1959 for continuous conduction mode is:

$$L = (V_{OUT} \times (V_{IN} - V_{OUT})) / (V_{IN} \times \Delta IL \times f)$$

$$L = (3,3 \times (7,4 - 3,3)) / (7,4 \times 0,060 \times 500 \times 10^3)$$

$$L = (3,3 \times 4,1) / (7,4 \times 30\,000)$$

$$L = 13,53 / 222\,000$$

$$L_{min} = 60,9 \mu H$$

Standard value: $L1 = 10 \mu H$ (standard serie, available with $I_{sat} > 500$ mA).

Note: with $L = 10 \mu H$, the actual ripple will be lower than the 30% target:

$$\Delta IL_{réel} = (V_{OUT} \times (V_{IN} - V_{OUT})) / (V_{IN} \times L \times f)$$

$$\Delta IL_{réel} = (3,3 \times 4,1) / (7,4 \times 10 \times 10^{-6} \times 500 \times 10^3)$$

$$\Delta IL_{real} \approx 365 \text{ mA} \rightarrow 18 \% \text{ ok IS } \checkmark (< 30 \%)$$

3. peak and valley currents in the inductor

$$I_{L_peak} = IS + \Delta IL_{real}/2 = 200 + 183 = 383 \text{ mA}$$

$$I_{L_valley} = IS - \Delta IL_{real}/2 = 200 - 183 = 17 \text{ mA}$$

Continuous conduction mode check: $I_{L_valley} > 0 \checkmark$

Saturation current required: $I_{sat} > I_{L_peak} = 383 \text{ mA} \rightarrow$ inductor standard $10 \mu H$ ok.

4. Output capacitor C3

4.1 ESR constraint (voltage ripple)

The output voltage ripple is mainly determined by the capacitor's ESR:

$$\begin{aligned}\Delta V_S &\approx \Delta I_{L_real} \times ESR \\ ESR_{max} &= \Delta V_S / \Delta I_{L_real} = 50 \times 10^{-3} / 0,365 \\ ESR_{max} &= 136 \text{ m}\Omega\end{aligned}$$

Chosen capacitor: AVX TPS 100 μ F / 10 V (tantalum solide, ESR typ. 100 m Ω < 136 m Ω) ✓

4.2 Capacitance value

The value in μ F is not critical. With C3 = 100 μ F:

$$\begin{aligned}\text{Capacitive ripple} &= \Delta I_L / (8 \times f \times C3) \\ &= 0,365 / (8 \times 500 \times 10^3 \times 100 \times 10^{-6}) \\ &= 0,91 \text{ mV (negligible compared to the ESR contribution)}\end{aligned}$$

5. Input capacitor C1

The RMS current in the input capacitor is:

$$\begin{aligned}I_{C1_rms} &= I_S \times \sqrt{(V_{OUT} \times (V_{IN} - V_{OUT})) / V_{IN}} \\ I_{C1_rms} &= 0,200 \times \sqrt{(3,3 \times 4,1 / 7,4)} \\ I_{C1_rms} &= 0,200 \times \sqrt{0,183} \\ I_{C1_rms} &\approx 85,5 \text{ mA}\end{aligned}$$

Chosen capacitor: C1 = 100 μ F / 10 V, tantalum AVX TPS (Iripple $\geq$ 0,7 A) ✓

6. Freewheel diode D1

Average current in D1 :

$$\begin{aligned}ID1_{avg} &= I_S \times (V_{IN} - V_{OUT}) / V_{IN} \\ ID1_{avg} &= 0,200 \times (7,4 - 3,3) / 7,4 \\ ID1_{avg} &= 110 \text{ mA}\end{aligned}$$

Max reverse voltage = VIN = 7,4 V

Selected Diode: MBRS330T3 (Schottky, 3 A / 30 V, VF typ. 0,5 V @ 3 A) ✓

7. Capacitor BOOST C2

The capacitor BOOST supplies the switch NPN driver to saturate it. The minimum value is:

$$\begin{aligned}C2_{min} &= (I_S / 50) / (f \times 0,6) \\ C2_{min} &= (0,200 / 50) / (500 \times 10^3 \times 0,6) \\ C2_{min} &\approx 13 \text{ nF} \rightarrow C2 \text{ selected} = 0,68 \text{ }\mu\text{F (datasheet standard value)} \quad \checkmark\end{aligned}$$

Diode D2 : 1N914 or 1N4148, anode connected to VOUT.

8. Feedback network (R1, R2)

The internal voltage is VREF = 1,21 V. The resistive divider R1/R2 fixes VOUT :

$$\begin{aligned}V_{OUT} &= V_{REF} \times (1 + R1/R2) \\ \rightarrow R1/R2 &= (V_{OUT}/V_{REF}) - 1 = (3,3/1,21) - 1 = 1,727\end{aligned}$$

Choice of R2 $\leq$ 2,5 k Ω (datasheet recommandation) : R2 = 2,49 k Ω

$$R1 = R2 \times (V_{OUT}/V_{REF} - 1) = 2490 \times 1,727$$

12. Simulated waveforms

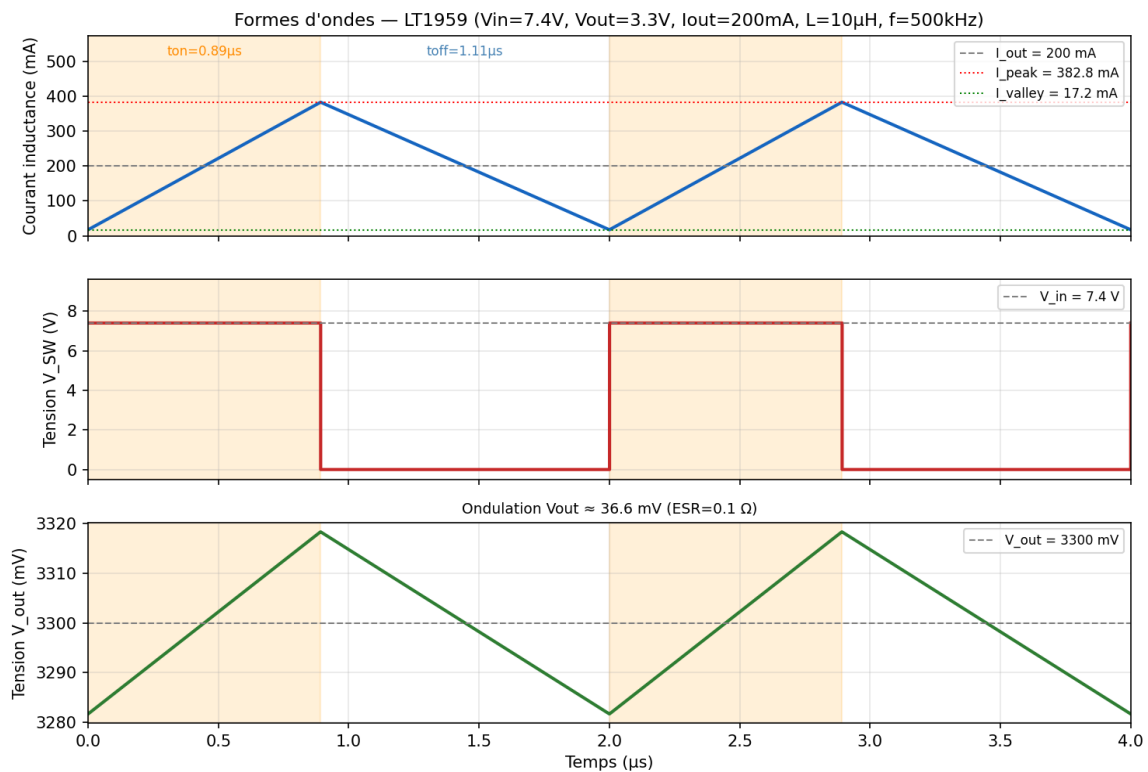


Figure 2 — Inductor current IL, voltage VSW and output voltage VOUT over 2 periods (f = 500 kHz)

The orange area represents the conduction time of the switch (ton ≈ 0,892 μs).

VOUT ripple is mainly due to the ESR of the capacitor C3 (≈ 36 mV < 50 mV spec) ✓

Reference : Linear Technology — LT1959 Datasheet, Rev. A.

Calculus realized for IS = 200 mA nominal. Check thermal parameters for higher currents.